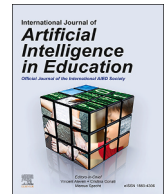




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Artificial intelligence to help people with dyslexia in education: An interdisciplinary literature review

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ABSTRACT

The rapid evolution of AI has reshaped many educational practices, raising new questions about its role in supporting students with learning disorders. This work presents a systematic and interdisciplinary review on the use of artificial intelligence (AI) and generative AI (GAI) to support students with dyslexia in educational settings. Dyslexia is the most common disorder among people and, despite the rapid development of AI technologies in recent years, the connections between AI, dyslexia, and education remain underexplored, and comprehensive, up-to-date analyses are lacking. To address this gap, an extensive literature, published between 2018 and 2024, was analyzed, ultimately identifying $n = 72$ relevant studies, guided by the PRISMA methodology. These studies were classified according to a set of semantic categories carefully defined by the authors, allowing to map current research trends and identify emerging interdisciplinary research directions. This analysis shows that AI has been used to enhance detection, assistive support, and personalized learning for students with dyslexia, yet the evidence remains fragmented and experimental validation is limited. While GAI offers promising opportunities for content generation and interactive support, its presence in the current literature is still minimal. The review highlights existing achievements, identifies key challenges, and outlines future research avenues to advance inclusive, AI-supported learning for individuals with dyslexia.

1. Introduction

In recent years, the rapid advancement and widespread adoption of Artificial Intelligence (AI) and GAI technologies have profoundly transformed the educational landscape. Machine learning (ML), Natural Language Processing, and Intelligent Tutoring Systems are increasingly being used to enhance teaching and learning processes through adaptive feedback, personalized content, and data-driven insights. These innovations have also influenced the evolution of online learning environments, including Massive Open Online Courses (MOOCs) (Kara, 2021), which have progressively integrated AI-based tools to improve learner engagement, assessment, and accessibility. Within this broader transformation, AI has emerged not only as a technological enabler but also as a catalyst for more inclusive and flexible educational practices. This is especially relevant for supporting learners with specific educational needs, such as those diagnosed with Specific Learning Disorders (SLD), including dyslexia, dysorthographia, and

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dyscalculia, who often require personalized feedback and personalized learning paths. The lack of appropriate educational tools and customized pedagogical strategies can exacerbate their difficulties, both in traditional and digitally mediated learning settings. The widespread occurrence of SLD makes this issue particularly significant. According to the European Dyslexia Association (EDA), epidemiological estimates in European countries range between 5% and 12% (American et al., 2000) of the population. The latest revision of the DSM-5 increased prevalence estimates to 5%-15% (American et al., 2013; Mazefsky et al., 2013), with dyslexia being the most common of all SLDs. As the EDA notes, “No matter which country – No matter which language – Dyslexia is everywhere.”

1.1. Motivations

Among assistive technologies designed to support students with dyslexia, those developed within TEL environments remain among the most widely adopted (Caute et al., 2018; Richardson & Lyytinen, 2014). Within TEL, AI has played a central role for many years, underpinning systems ranging from early ITS to recent generations of GAI-driven conversational agents such as ChatGPT.² These developments illustrate how AI has progressively evolved into a key component of personalized and accessible learning support.

In this paper, we present an in-depth analysis of interdisciplinary literature concerning works addressing the intersection of the following three fields of interest: AI, dyslexia, and education, starting from 2018 (when transformers were proposed) to October 2024. While this review covers both AI-driven diagnostic approaches and AI-based assistive or educational tools for individuals with dyslexia, these two domains are not treated interchangeably. In contrast, they are considered as distinct research strands within our analytical framework. Their joint inclusion is intentional, as both contribute to understanding how AI supports learners with dyslexia before and after diagnosis in educational contexts. The rationale of this study in the context of existing knowledge is, first, to reflect on and to give useful indications about the state of the art guided by some interdisciplinary categories identified by the authors, and, second, to show future trends and open challenges. Our study seeks to fill a gap in the literature, noting that, to our knowledge, the mixed area: AI, education and dyslexia has never been thoroughly examined. In fact, the reviews that have been published to date do not comprehensively cover the intersection of these three areas, especially in the use of GAI.

Following the above guidelines, we can summarize our study into the following two research questions:

1.2. RQ1

What does the current body of research reveal about how AI is being used to support students with dyslexia in educational settings?

A thorough literature review was conducted, based on the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) method (Page et al., 2021). A large number of papers were analyzed, first selecting more than $n = 500$ papers and then classifying them through an exhaustive and interdisciplinary content analysis. Finally, $n = 72$ relevant articles were reported and classified.

1.3. RQ2

What challenges and future research directions emerge from recent applications of Artificial Intelligence aimed at assisting learners with dyslexia?

Our results show that, at present, the contribution of AI and GAI to students with dyslexia is minimal, and, while experimental methods are mentioned, the field lacks properly designed experimental studies. This study is a more complete, in-depth, and updated version of the short review on the same topics, presented in (Dabaghi et al., 2024).

1.4. Paper organization

The paper is organized as follows. Section 2 presents an overview of Dyslexia. Section 3 discusses the relationship between AI and Dyslexia before 2018. Section 4 describes the PRISMA method followed to select relevant papers, while in section 5 the results of the retrieval step are presented. In section 6 we take a deep dive into the results that respond to RQ1, giving some useful indications about the relevance of this review to the field. In section 7, we respond to RQ2 by presenting open challenges and future trends. Finally, conclusions and plans for future studies are presented in section 8.

2. Dyslexia

Dyslexia is a widespread form of learning difficulty, with reported prevalence estimates typically ranging between 5% and 10%, depending on diagnostic criteria, linguistic context, and assessment methods (Knight, 2018; Shaywitz, 2012; Siegel, 2006). It may be the disability that students in higher education self-identify most frequently (Mortimore & Crozier, 2006), which represents neurobiological processing problems that slow the process of learning while acquiring knowledge across languages and are characterized by difficulties in acquiring the skills to recognize words, spelling, and writing (El Kah & Lakhouaja, 2018; Leij, 2013; Ruberto et al., 2016; Tunmer & Greaney, 2010). In particular, while children with dyslexia often have difficulty with word-reading accuracy, older

² www.openai.com.

adolescents and adults typically have more pronounced difficulties reading fluency (speed and automaticity), even when accuracy has partially improved (Christodoulou et al., 2014; Reis et al., 2020).

It is a complex neurobiological learning disorder, so numbers evaluating its prevalence remain hypothetical or rough estimates. Screening for this disorder can vary greatly depending on regions and culture, language, and diagnostic methods (Kearns et al., 2019). Dyslexia is a brain-based genetic learning disability that persists throughout life; it is not an illness and therefore cannot be “cured,” but individuals can learn effective strategies to manage it.³ In fact, when diagnosed early and properly managed, it is possible to overcome many of the learning difficulties that result from it.

Across Europe, the diversity of languages and the demands of multiple languages may pose particular challenges to children and adults with disabilities, given their language learning difficulties. Intelligence, individual effort, or socioeconomic status are not factors that contribute to dyslexia (Aaron, 1997; Mortimore & Crozier, 2006; Tunmer & Greaney, 2010). Rather, it is cognitive difficulties that cause dyslexia, which can also affect verbal learning in other disciplines, including arithmetic (Simmons & Singleton, 2008; Vukovic et al., 2010). Furthermore, organizational skills and motor coordination difficulties can be present in students with dyslexia, but these are not central to the condition (Mortimore & Crozier, 2006).

Due to these situations, it becomes an ongoing challenge for people with dyslexia to navigate life,⁴ including difficulties handling small tasks, as well as larger tasks that involve mathematics, reading, and writing. Thus, educational facilities and teachers have the responsibility to obtain sufficient training to help students progress and improve their skills in these areas. Teachers certainly cannot address the neurobiological reason for the appearance of dyslexia, but can help the student apply effective learning techniques. Educational institutions can train teachers and keep them up to date with the latest research on the cognitive and behavioral aspects of dyslexia to ensure a strong and nuanced understanding of this disorder (Siegel, 2006). It is also important that they understand what schools are required to provide according to the law (Knight, 2018) or according to practices and policies (Mortimore & Crozier, 2006). Thus, through continuous assistance and targeted efforts, people suffering from dyslexia are capable of managing their challenges and achieving their educational and life goals.

3. AI for dyslexia before 2018

Before the introduction of Transformer models, AI solutions aimed at supporting students with dyslexia revolved mainly around assistive technologies, personalized learning environments, early detection strategies, and foundational NLP applications (Smith & Hattingh, 2020). Text-to-speech and speech-to-text systems were prominent tools, providing crucial support for reading comprehension and written communication (MacArthur, 2009), which has word recognition, spelling, and writing—the key problems encountered by people with dyslexia—as the necessary condition to acquiring these key life skills. In addition, ML facilitated the development of adaptive educational content tailored to individual learners, thus enhancing personalized educational experiences (Rose & Meyer, 2002). AI-driven diagnostic tools employing machine learning algorithms have also become prevalent, analyzing reading patterns and cognitive profiles to enable the early and precise identification of dyslexia (Kohli & Prasad, 2010). Moreover, neuroscience research used machine learning methods to gain deeper insight into the neurobiological foundations of dyslexia (Vandermosten et al., 2016). However, there were limitations to the use of the aforementioned AI techniques. Early AI systems struggled to adapt effectively to individual learning differences due to limited complexity and reliance on rule-based or basic statistical methods. For example, in Text-to-Speech Technologies, previous systems lacked naturalness and expressivity, potentially reducing engagement and comprehension among learners with dyslexia, while AI diagnostic tools before 2018 lacked robustness, often relying heavily on limited datasets, which diminished their reliability and generability for diverse student populations (Khan et al., 2018). Finally, AI decision-making processes lacked transparency, making it difficult for educators and learners to understand, trust, or adequately assess the decisions and recommendations provided by AI tools (Selwyn, 2021). As noted in this study, the advent of transformers has enormously improved the possibility and effectiveness of supporting students with dyslexia, thus opening new frontiers of research, addressed in this study.

We selected 2018 as the starting year because it coincides with the emergence of Transformer architectures and the onset of the modern deep learning era. This transition marks a qualitative leap in the capabilities of AI applied to education and accessibility, allowing the creation of more adaptive and linguistically powerful systems. Thus, 2018 represents a natural boundary between traditional AI tools and the new generation of GAI-enabled approaches explored in this review.

4. Methods

In this section, we describe the process of collecting and analyzing relevant articles from the literature. As highlighted in the Introduction, we adhered to the PRISMA method, consisting of three macro-phases: Identification, Screening, and Inclusion (Page et al., 2021). The next paragraphs detail each step of the method as applied to our study. Fig. 1 shows the entire PRISMA workflow followed as our baseline. For a better understanding, of the PRISMA workflow, a *Report* could be a journal article, a preprint, a conference abstract, or any other document providing relevant information while a *Record* could be the title or abstract (or both) of a report indexed in a database or website.

³ Dyslexia Information Page. U.S. National Institutes of Health: <https://www.ninds.nih.gov/health-information/disorders/dyslexia>.

⁴ <https://eda-info.eu/what-is-dyslexia/>.

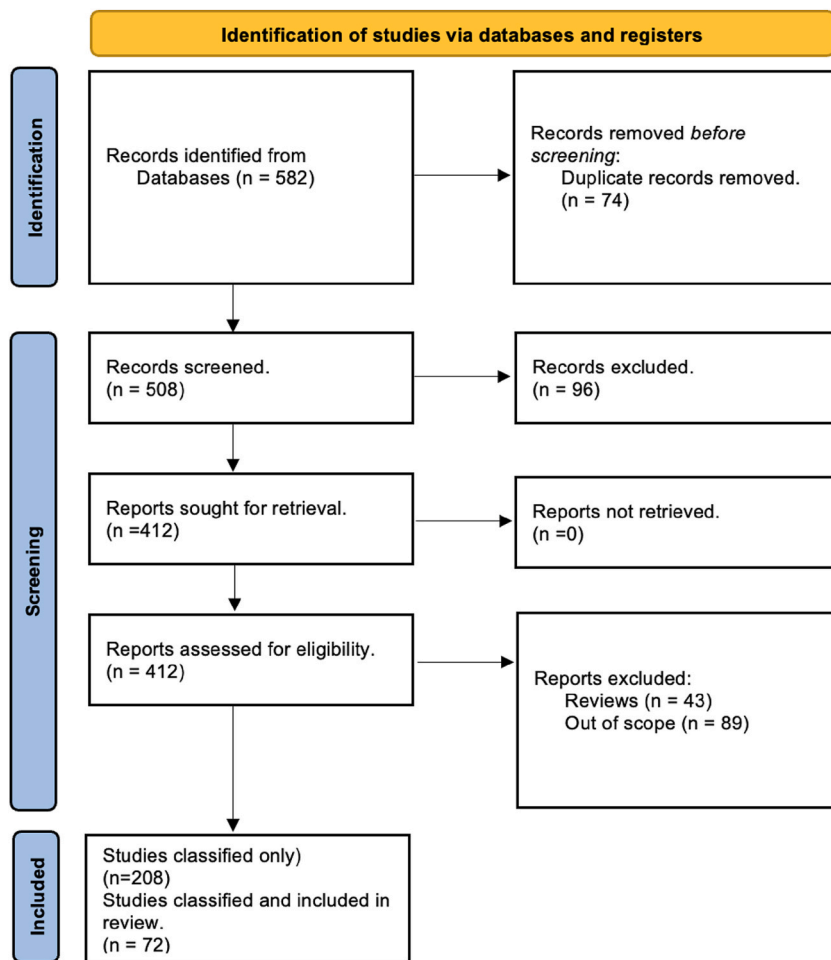


Fig. 1. The whole PRISMA workflow followed in the study.

4.1. Step 1: The identification phase

In this step, a set of papers is first identified from selected sources and a subset of them removed, based on duplicates removal.

4.1.1. The information sources

For the review, the search engines shown in Table 1 were used.

4.1.2. The search strategy

Due to our focus on dyslexia as the most prominent SLD, and to select and retrieve relevant articles situated at the intersection of AI, dyslexia, and education, we used the following query for all the search engines of Table 1:

“artificial intelligence AND dyslexia AND education”

Table 1

The retrieval engines used for searching papers. The Elsevier search engine included the Scopus database.

Source Database	Link to the Search Engine
IEEE	https://ieeexplore.ieee.org/Xplore/home.jsp
Elsevier	https://www.sciencedirect.com/
Springer	https://link.springer.com/
Google Scholar	https://scholar.google.com/
Arxiv	https://arxiv.org/
ACM	https://dl.acm.org/

with the main goal of discussing papers that focus on the use of AI to help people with dyslexia improve their learning process. Each retrieved paper had a group of related meta-data, each publisher having its own paper-tagging method. In order to have a uniform set of information, the different tags were normalized by hand. The search identified $n = 582$ articles in the source databases of Table 1. Subsequently, a set of $n = 74$ duplicated records was eliminated. At the end of the identification process, we had a set of $n = 508$ papers for screening.

4.2. Step 2: The selection phase

In this step, the entire set of identified articles was selected, following the PRISMA guidelines: first the set of records screened, second the set of reports sought for retrieval, and finally the set of records assessed for eligibility (Page et al., 2021).

4.2.1. Record screened

Taking into account the rationale discussed in Sect. 1, we considered as eligible papers the papers that met the eligibility criteria shown in Table 2.

Consequently, we excluded $n = 96$ articles, including those sourced from medical reports, non-official sources, or those not written in English, or not written in the time interval considered for the review. As shown in Table 3, most of the papers came from the Springer database. The distribution of the screened papers is shown in Table 4: most of the papers were conference papers, followed by a good number of articles from journals and reviews, and very few books and book chapters on the topic.

4.2.2. Reports sought for retrieval: semantic categories

In sum, $n = 412$ papers were retrieved, so no records were left unretrieved ($n = 0$). As the first result of the analysis of all the retrieved articles, several semantic categories were identified by which to classify all the retrieved articles. Table 5 shows the categories. To the *Out of Scope* category, not shown in the table, belong those papers that did not match the eligibility criteria. We propose these categories for a better synthesis and analysis of the literature from an interdisciplinary point of view, discussed in the next section. In fact, many papers belonged to more than one category.

4.2.3. Reports assessed for eligibility

The authors determined inclusion in this review by conducting an iterative group and individual work. In a first round, we eliminated papers that were irrelevant because the search term produced a non-respondent result. The papers were then divided into thirds, and each author individually eliminated additional irrelevant papers based on close reading of the papers themselves.

Subsequently, we individually categorized the papers and then met to agree on the categorization. The coding activity was supported by a shared digital environment that allowed all authors to assign and cross-check semantic tags collaboratively. This technological support and the detailed workflow of the tagging process are described later in the paper. Each author then reviewed the coding, ensuring agreement of the assigned codes based on the group discussion.

Another session of all authors allowed for agreement on the description of the final categories. In total, we excluded $n = 132$ articles: $n = 89$ *Out of Scope* papers, and $n = 43$ reviews, leaving $n = 280$. As illustrated in Fig. 1, of the 280 papers, only $n = 72$ remained for the final analysis, all papers that displayed the interdisciplinarity of the intersection of our search topics. We used a consensus approach to inter-rater reliability through this iterative process.

4.2.4. Review papers analysis

Although we had retrieved a set of $n = 43$ review articles, we decided not to use them for the study because we wanted to go directly to the source articles rather than papers mediated by others. However, Table 6 shows the articles dealing with topics closely related to our survey, while Table 7 shows the temporal distribution of all the reviews retrieved.

In fact, of all the articles retrieved and classified as review articles, only eight are tangentially related to our mixed area of investigation, confirming a lack of comprehensive surveys focused specifically on the intersection between AI and dyslexia. Most of these works adopt a broad perspective on learning disabilities, six out of eight address learning disorders in general, and therefore fail to provide a focused synthesis of technologies related to dyslexia. Two reviews, namely (Zhai & Krajcik, 2023) and (Aliu, 2024), are the most relevant to our topic, as they explore the use of AI for learning disabilities. However, their scope remains limited in both methodological depth and temporal coverage. In particular, none of the retrieved articles, classified as review articles, address the recent impact of GAI, such as large language models, image generators, or multi-modal tools, on accessibility, diagnosis, and personalized learning for people with dyslexia. Our work fills this gap by mapping the literature in light of the GAI revolution, offering

Table 2
The eligibility criteria.

Criterion	Value
Language	English
Period	2018-2024 (October)
Paper type	Journal, Conference, Book, Book Chapter, Review
Focus	Artificial Intelligence to help students with dyslexia
Query	artificial intelligence AND dyslexia AND education

Table 3
The distribution of the retrieved papers by source.

Source Database	# Eligible papers
IEEE	38
Elsevier	59
Springer	258
ACM	14
Other sources (Arxiv, Scholar, MDPI,CRC,SAGE, ...)	43
Total	412

Table 4
Distribution of the Papers by type.

Type	#
Journal	140
Conference	221
Book	2
Book Chapter	6
Review	43
	412

Table 5
The proposed semantic categories.

Category	Description
Research	Articles primarily engaged in research discussions on the theme
Help	Articles centered on the development and usage of assistive or educational tools for learners with dyslexia (distinct from diagnostic approaches).
Medical	Articles with a medical orientation, exploring the physiological aspects of dyslexia
Review	Articles offering a review of existing literature and studies on the topic
Education	Articles reporting on experimentation or application in the educational field: school or university level
Game	Articles exploring gamification approaches to support individuals with dyslexia
Detection	Articles focused on innovative methods and techniques to detect dyslexia (diagnostic applications; conceptually distinct from pedagogical or assistive tools).

Table 6
The review papers of interest for the study.

Year	Ref.	Description
2020	Smith and Hattingh (2020)	Assistive technologies available to support students with Dyslexia
2020	Kaisar (2020)	Developmental dyslexia detection using machine learning techniques
2021	Lerga et al. (2021)	An overview on the use of assistive technologies in education with emphasis on students with dyslexia
2021	Dobson Waters and Torgerson (2021)	Studies to evaluate interventions aimed at improving outcomes for students with dyslexia
2022	Barua et al. (2022)	AI-assisted tools, developed using machine learning models, to address learning challenges in students with dyslexia
2022	Stanley et al. (2022)	A survey on existing chatbot accessibility guidance: how chatbots can deliver a successful experience for all users.
2023	Mcvey et al. (2022)	Educational Robotics and Dyslexia
2023	Zhai and Krajcik (2023)	How artificial intelligence (AI) has been employed to support students with learning disabilities
2024	Aliu (2024)	A review about the use of AI in special education in general

Table 7
The temporal distribution of the retrieved review papers between 2018 and 2024 (October).

Year	# Review
2024	4
2023	12
2022	12
2021	6
2020	5
2019	2
2018	2
	43

an updated analysis of how these emerging technologies are being used, evaluated, and discussed within the context of dyslexia.

5. Results

The process of identifying, screening, and including articles, as depicted in Fig. 1, was described in the previous section. In the upcoming sections, we present some quantitative and qualitative results.

It is important to note that the purpose of this review is to map and analyze the scientific landscape concerning the intersection of Artificial Intelligence, dyslexia, and education. Consequently, we focus on identifying trends, research objectives, methodological approaches, and the main AI paradigms used, rather than evaluating or comparing individual applications. This perspective allows for a broader and more inclusive understanding of how AI has been applied to dyslexia over time, highlighting the evolution of research directions and emerging areas of interest.

5.1. Analysis of the eligible papers

Fig. 2 shows the steps of the process that produce, for each category, the list of all its (tagged) papers. As a first step, we analyze each paper. As a second step, the semantic categories were fixed. After that, each paper was initially manually tagged with one or more categories among those of interest (see Sect. 4). Finally, by a Python-based procedure all the tags of the papers were processed, producing for each category the list of its papers. Notice that a paper could belong to more than one category.

The coding process was supported by a shared Google Worksheet, where each author manually assigned one or more semantic tags to each eligible paper according to the predefined category framework (see Sect. 4). This spreadsheet served as the primary coding tool. Each row corresponded to a paper, while the columns represented categories such as *assistive technologies*, *diagnosis and assessment*, *learning support*, *teacher training*, and *policy and inclusion*.

To ensure consistency and reproducibility, a simple Python script was then used to merge and process the tagged data exported from the spreadsheet. The program aggregated all tags and generated the final category lists reported in Figs. 3 and 4. This semi-automated approach facilitated cross-checking among authors while maintaining full transparency in the tagging rationale.

5.1.1. Papers distribution by semantic categories

The results of the tagging process are summarized in Figs. 3 and 4. The first figure shows the distribution of the retrieved papers belonging to one category only ($n = 341$), while the second figure shows the distribution of the papers classified into two categories.

5.1.2. Papers distribution

To ensure methodological consistency, we adopted a multi-stage, consensus-based approach to paper selection and categorization. After independent selection and iterative discussions among all authors, we identified several semantic categories reflecting different conceptual areas. For in-depth analysis, we focused on articles located at the intersection of two or more categories. This choice was motivated by our research interest in interdisciplinarity within the educational field: papers appearing in intersecting sets are those in which multiple conceptual dimensions coexist, thus representing the most relevant examples of integrative and cross-domain research in our dataset.

Figs. 5–7 show the complete picture of both the distribution of the number of items in the individual categories and their distribution in the intersecting sets.

6. Supporting students with dyslexia in educational settings using AI (RQ1)

In this section, we answer the research question RQ1, delving deeper into the findings presented earlier. Here we focus on interpreting the implications of these findings, as all descriptive and quantitative results have been reported exclusively in Section 5. It is important to note that AI for dyslexia detection and AI for educational or assistive support represent different research domains with distinct objectives and methodologies. In our review, these domains are not conflated, but analyzed as complementary components of the broader interdisciplinary landscape shaped by AI, dyslexia, and education. Their separation is reflected in our semantic categories (e.g., Detection vs. Help/Education) and in the intersection analysis presented earlier. Moreover, it is important to acknowledge the limitations of the study, such as potential biases in the literature search and selection process, as well as the exclusion of non-English language studies, which could have offered additional insight. Our analysis is based on the different semantic categories used to

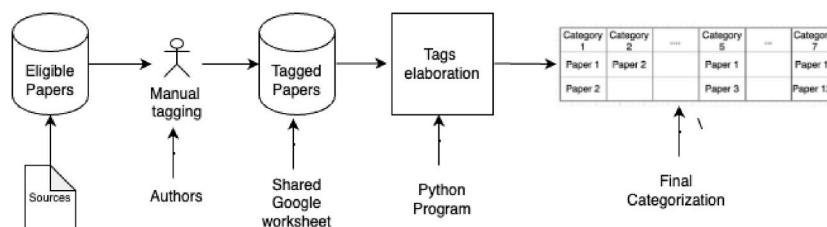


Fig. 2. The workflow producing categorized papers.

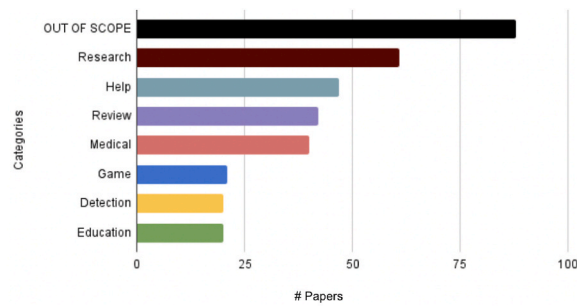


Fig. 3. The distribution of the retrieved papers belonging to one category only ($n = 341$).

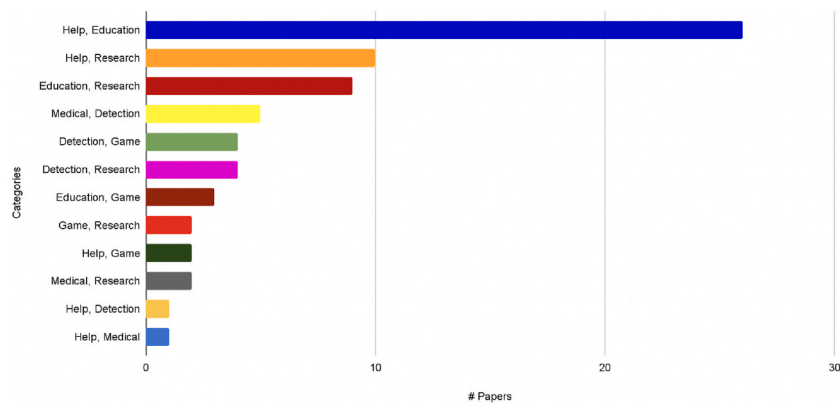


Fig. 4. The distribution of the papers belonging to two categories, excluding the Reviews and the Out-of-Scope papers.

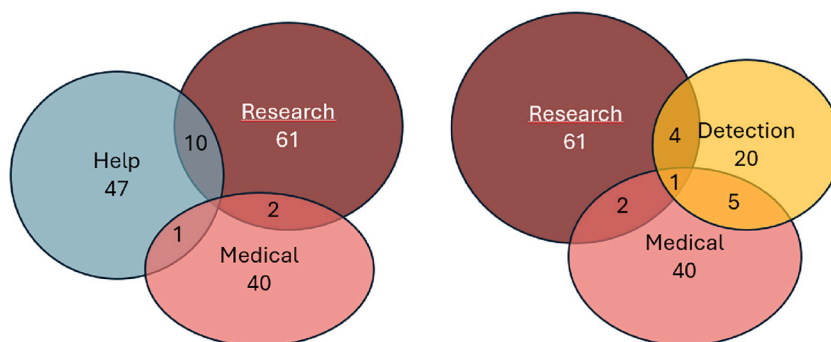


Fig. 5. The intersections among the categories: Research-Help-Medical (left) and the categories: Research-Medical-Detection (right), having respectively zero and one paper in common among the sets. The number under the name of the category is for the number of papers belonging to that category only.

classify the retrieved articles, and as the relevant papers, we consider the papers laying at the intersection of two categories at least, for the reasons explained in the previous section.

Taken together, the reviewed studies depict a research landscape that is rapidly expanding but structurally fragmented. While AI applications for dyslexia span detection, assistive technologies, educational support, and generative systems, these strands often evolve in parallel rather than in integration. Technological sophistication, particularly in detection models and recent generative approaches, does not always correspond to pedagogical embedding or validated instructional impact. This imbalance suggests that the field is currently driven more by technological opportunity than by consolidated educational theory, highlighting the need for closer alignment between AI development, cognitive frameworks, and classroom practice.

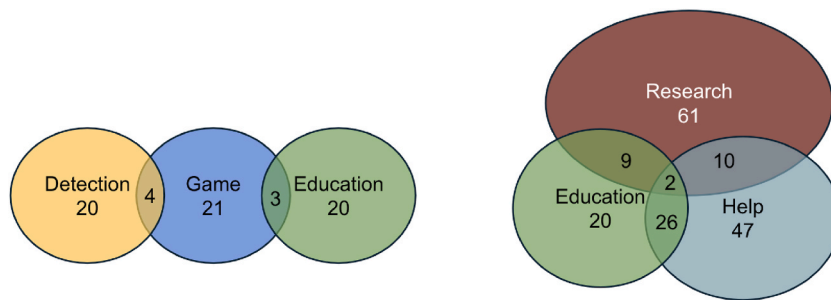


Fig. 6. The intersections among the categories: Game-Education-Detection (left) and the categories: Education-Help-Research (right), having respectively zero and two papers in common among the sets. The number under the name of the category is for the number of papers belonging to that category only.

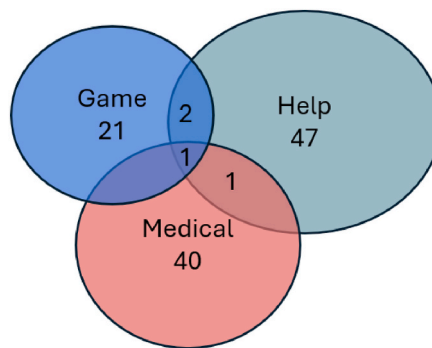


Fig. 7. The intersections among the categories: Game-Help-Medical, having one paper in common. The number under the name of the category is for the number of papers belonging to that category only.

6.1. Help-education

In this section, we want to provide useful information on the items retrieved that belong to the intersection of the two categories Help and Education. This is because this set represents, with its numerosity, the most significant set in our survey. Specifically, we divide the set into two subsets: articles dealing with ML and those dealing with GAI, shown, respectively, in Table 8 and Table 9. The reasons why we separate the articles into these two subsets stem from the fact that since the birth of GAI, the artificial intelligence techniques used in this field have split into two approaches, even though in the end GAI is a subset of ML. Moreover, as can be seen by looking at the tables above, the number of articles using GAI in 2024 is significantly higher than that using classical ML techniques. This means that GAI is now taking over classical ML as a tool to help the learner. This phase is characterized by the emergence of GAI technologies that have been revolutionizing a large number of research fields starting from 30 November 2022, when ChatGPT was released by OpenAI. However, it should be noted that there is a notable dearth of substantial research exploring the intersection of GAI

Table 8

ML for helping students with dyslexia.

Year	Ref.	Description
2018	Benmarrakchi et al. (2018)	Presents a student model for personalizing students with disabilities.
2019	Della Ventura (2019)	Presents an AI-based software to enhance the skills of the students with dyslexia in music
2020	Hauwaert et al. (2020)	An app is presented to enhance learning by an educational adventure game
2021	Wang et al. (2021)	An AI-based Augmentative Alternative Communication for student with dyslexia
2021	Tayal (2021)	Deep Learning to teach numbers to students with dyslexia
2022	Mcvey et al. (2022)	Reinforcement Learning in Robotics to Support Students with dyslexia
2022	Sharif and Elmedany (2022)	ML to identify difficulty levels for students with learning difficulties
2022	Mateos-Sanchez et al. (2022)	A Chatbot, as educational tool for people with intellectual disabilities
2022	Shardlow et al. (2022)	A software designed to facilitate the engagement of university students
2022	Guo et al. (2023)	A ML-based practice with assistive pen for young learners with dyslexia
2023	Gilbert et al. (2023)	AI to identify and create font to improve reading ability of people with dyslexia
2023	Mitra et al. (2023)	Data analysis and ML for improving education for students with special needs
2024	Zingoni et al. (2024)	A ML-based predictor to support university students with dyslexia
2024	Kaluri et al. (2024)	A book on implementing inclusive open practices
2024	Morciano et al. (2024)	Using AI to suggest the most suitable supporting tools for students with dyslexia

Table 9

Help systems based on GAI.

Year	Ref.	Description
2024	Li (2024)	- An AI-powered reading augmentation tool based on ChatGPT and DALL-e
2024	D'Urso and Sciarone (2024)	- An intelligent chatbot capable of analyzing conversational data to model students' understandings and misconceptions, thereby offering personalized educational support
		- Web application
		- Contextual Concept Maps generation
		- ChatGPT based (RAG and Function Calling)
		- Content creation
2024	De Marco et al. (2024)	- Intelligent chatbot
		- Web application
		- Multimedia Concept Maps generator
		- Tailored on students with dyslexia
		- ChatGPT based
2024	Khazanchi et al. (2024)	- Provides a comprehensive overview of the potential of generative AI tools to improve the preparation of teachers in the context of inclusive education
2024	Gilbert et al. (2023)	- Discusses important challenges for Special Education Teachers about GAI and teaching strategies for students with special needs
2024	Bhuvan et al. (2024)	- Discusses several ways in which ChatGPT can aid people with dyslexia.
2024	Subramani et al. (2024)	- An Intelligent chatbot that can interpret and produce voice-based interactions in addition to understanding and responding to textual queries
		- ChatGPT, NLP, GAN
		- The chatbot appears as a virtual person and creating an immersive atmosphere
2024	Goodman et al. (2024)	- LamPost: an LLM-enabled prototype to support adults with dyslexia with e-mail writing
2024	Habib et al. (2024)	The impact of ChatGPT-3 on students' divergent thinking in a college-level creativity course
2024	Mayilyan (2024)	- Explores the integration of ChatGPT into STEM education as a means to address educational disparities and enhance learning experiences.
2024	Grover (2024)	- Explores the intersection of GAI and education, examining the applications, benefits, challenges, and ethical considerations of integrating GAI into educational practices
		- Provides a taxonomy of possible aids of GAI, including for students with dyslexia

with educational strategies for individuals with dyslexia and in general with special needs.

Help and education-oriented studies collectively show that AI tools offer localized benefits but remain weakly integrated into structured pedagogies. Their impact is fragmented by heterogeneous evaluation methods and a lack of longitudinal evidence. A coherent synthesis suggests the need for ecosystem-level approaches where AI tools are embedded within established instructional models rather than used as isolated prototypes.

6.1.1. Machine learning

This category, as the intersection of *Help* and *Education*, has, as illustrated in Table 5, articles centered on the development and use of tools aimed at assisting learners with dyslexia, reporting on experimentation or applications in the educational field, be it at the school or university level.

The articles in Table 8 show a fairly diverse overview of the literature. In particular, we have identified the following five areas of research. The first is specific applications. To this area belong papers using ML to improve learning in particular knowledge domains. The articles (Della Ventura, 2019; Tayal, 2021) present ML techniques to help students with dyslexia to learn in two particular knowledge domains: music and numbers. The second is engagement. To this area belong the papers using ML to improve students' engagement (Shardlow et al., 2022). The third is personalization. ML for the personalization of students with learning disabilities is the main focus of this area (Benmarrakchi et al., 2018). The fourth is recommendation. This area includes articles proposing the use of ML to recommend tools to students with dyslexia (Morciano et al., 2024; Sharif & Elmedany, 2022; Zingoni et al., 2024). Finally, the fifth is generic support. This category consists of proposals that use ML as a general support for learning (Gilbert et al., 2023; Guo et al., 2023; Hauwaert et al., 2020; Kaluri et al., 2024; Mateos-Sanchez et al., 2022; Mcvey et al., 2022; Mitra et al., 2023; Wang et al., 2021).

ML studies emphasize technical performance, feature extraction, and classification accuracy, while overlooking ecological validity and practical deployment. High accuracy in controlled settings rarely translates into classroom use. A cross-study analysis indicates that future work should combine technical metrics with learner centered indicators such as usability and instructional relevance.

6.1.2. Generative AI

The capabilities of transformer-based AI models to understand and generate human-like text present a myriad of potential applications in education. These range from personalized tutoring systems that adapt to individual learning to intelligent content generation tools that assist in creating customized learning materials.

The analysis of these papers pointed out the following three areas of research. First, intelligent chatbots. This area of research covers proposals using conversational agents to interact with students with dyslexia to help improve the learning process (Li, 2024; D'Urso & Sciarone, 2024; Subramani et al., 2024; De Marco et al., 2024). The second is teacher training. This area of research addresses the GAI for teachers to learn how to use it (Gilbert et al., 2023; Khazanchi et al., 2024). And the third is exploratory studies. This area addresses exploratory studies on the use of GAI in education (Bhuvan et al., 2024; Goodman et al., 2024; Grover, 2024; Habib

et al., 2024; Mayilyan, 2024). All articles in these three categories are from 2024, which indicates the growth in interest in the use of GAI in education, at various levels, over the past year.

GAI research spans assistive tools, conversational agents, and material generation, revealing conceptual fragmentation but a shared potential to support meta-cognitive processes. Synthesizing these perspectives highlights the need for clearer theoretical models explaining how GAI influences cognitive load, engagement, and self-regulation in learners with dyslexia.

From a practical perspective, these findings suggest that AI-based tools for learners with dyslexia should not be implemented as isolated technological solutions, but rather embedded within structured pedagogical frameworks. Educators and instructional designers need to align AI functionalities with established educational principles, such as differentiated instruction and cognitive load management. For researchers, the results indicate the importance of moving beyond prototype validation toward longitudinal classroom-based studies that assess not only technical performance but also learning outcomes, usability, and sustained engagement. Greater collaboration between AI developers, special education experts, and classroom practitioners appears essential to ensure ecological validity and real-world applicability.

6.2. Other sets of papers

The analysis of the relevant papers ($n = 46$), each of them accompanied by a brief description, is shown in Table 10, Table 11 and Table 12. Additional paper sets show uneven development across domains—technically advanced detection studies contrast with pedagogically oriented but technically limited game-based work. This pattern suggests disciplinary silos and underscores the need for integrative designs that align technical innovation with validated learning theories.

Fig. 8 reports the papers' distribution per year. The relevant papers show a significant increase in interest in the literature in the last two years. Some general reflections on the whole set of items covered in the three tables:

From the analysis of the tables summarizing the other set of papers, we can make several observations about the topics, quality, and distribution of the related papers. To start with, the studies reveal a broad application of AI technologies, from personalized educational aids to advanced diagnostic tools. This diversity underscores the potential of AI to provide personalized support for students and scalable solutions to detect dyslexia in educational settings. Many studies, particularly those classified under the Detection and Medical categories, emphasize the development of early diagnostic tools. These tools frequently leverage machine learning, EEG, and eye-tracking technologies to improve accuracy and accessibility in the detection of dyslexia, reflecting the priority of early intervention. The Help and Education categories focus on AI-driven support tools designed to tailor educational content and adapt learning experiences to individual students' needs. These studies aim to make education more accessible to students with dyslexia - through adaptive learning systems, personalized recommendations, and digital platforms that adjust to the learning pace and style of each student. A notable subset of studies applies gamification techniques to motivate students with dyslexia, making learning more

Table 10
The other sets of papers.

Category	Ref.	Description
Detection-Medical	Gupte et al. (2023)	A ML model to analyze EEG signals from people with learning difficulties and provide results in minutes with the highest level of accuracy
	Ileri et al. (2022)	To diagnose dyslexia using EOG signals that are recorded simultaneously during reading texts, which are prepared in different typefaces and fonts
	Gellert and Elbro (2018)	The predictive validity of a dynamic test of decoding in which participants are taught three novel letters and how to synthesize the corresponding letter sounds into new words
	Abd Rahman and Ismail (2019)	An empirical study to determine a score on assessments of reading activities. A case study in a Malaysian school
	Subramanyam et al. (2019)	An interesting case study: a system to help improve the detection of dyscalculia
Research-Help	Wadhera and Mahmud (2022)	A computational approach to explore the influences of social learning on social cognition among individuals with Autism Spectrum Disorder
	Miranda Floriano et al. (2022)	Application of a new method focused on the accessibility evaluation of e-learning systems
	Quiñones (2020)	To identify, if changing the way a story is presented, influences the emotional impact of the user
	Alharbi (2022)	To examine creatively the viewpoints of individuals on distinct facial expressions and body language
	Lillywhite and Wolbring (2023)	How and to what extent does the AI/ML/robot literature include well-being in relation to disabled people
	Gilbert et al. (2023)	A project that seeks to use AI to design a font with the ability to help people with dyslexia read
	Minoofam et al. (2022)	This study introduces the RALF system to generate content automatically for students with dyslexia
	Rajapakse et al. (2018)	The results of an evaluation of a prototype mobile application which helps people with dyslexia to deal with their reading difficulties in real life successfully
	Kamarozaman et al. (2023)	A novel approach to assist students with dyslexia, ADHD, and short attention span in digesting any text-based information more efficiently.
	Tejero et al. (2020)	An experiment to test if complementary audio versions of the Video Messages Signs would be helpful for drivers with dyslexia
Help-Medical	Xu et al. (2021)	Intelligent learning environment and the intelligent education technology and application in different historical development stages for people with dyslexia
Game-Help-Medical	Khaleghi et al. (2022)	This study examines gamification and serious game approaches efficacy in improving motivation in children with disabilities.

Table 11
The other sets of papers.

Category	Ref.	Description
Detection-Game	Benalcázar Chicaiza et al. (2019)	This research aims to demonstrate the relationship between M-learning as a didactic strategy and children with dyslexia, in order to use innovative strategies that help in their development.
	Alluhaidan et al. (2024)	This research introduces a standardised approach utilising an online gamified test dataset to predict the risk of dyslexia through an artificial neural network model
Research-Education	BU et al. (2023)	This review summarizes the advantages of using machine learning for early screening of dyslexia.
	Kaisar and Chowdhury (2022)	This paper addresses the imbalanced dataset obtained from dyslexia pre-screening tests and proposes an oversampling and ensemble-based machine learning technique for the detection of dyslexia
	Praveena et al. (2024)	Provides an overview of the types and efficacy of AI-assisted tools created by machine learning models and deep learning models used to solve learning issues
	Lopez and Rangappa (2024)	What is the role of AI in early identification and intervention for students with learning disabilities (dyslexia)?
	Kharbat et al. (2021)	Highlights the gaps in incorporating artificial intelligence and its service to individuals with intellectual disabilities
	Ali et al. (2025)	This work aims to provide an understanding of the clinical procedures involved in diagnosing dyslexia alongside current interventions
	Wang et al. (2022)	Smart assistance to dyslexia students using artificial intelligence based augmentative alternative communication
	Ogunlade et al. (2023)	The study examines how students with dyslexia fare in their reading and spelling abilities after conversational AI structures are introduced as teaching tools in a Nigerian case study
	Nair et al. (2023)	This research examines the significance and effectiveness of AI-based reading intervention for students with dyslexia
	Ahmad et al. (2025)	AI as solution to the field of special education particularly in mainstream settings
	Paít et al. (2024)	AI to design a font with the ability to help people with dyslexia read

Table 12
The other sets of papers.

Category	Ref.	Description
Game-Education	Zavaraki (2024)	Opportunities of AI for people with special needs, in teaching and learning process, are introduced and analyzed
	Ouherrou et al. (2018)	Design and evaluation of an educational game developed for children with learning disabilities to assess their skills in Arabic language
	Jaramillo-Alcázar et al. (2021)	An interesting approach to the development of accessible serious games for children with dyslexia
Game-Help	Atkar and Jayaraju (2021)	The paper aims to use generative neural networks, for generating raw audio data for children recuperating from dyslexia
	Ma (2023)	This paper explores the introduction of VR games in education, with a focus on supporting students with dyslexia
Help-Education-Research	Chen and Perez (2023)	An assessment and instructional program, for assessing and facilitating the progress of language is discussed
	Mcvey et al. (2022)	The introduction of a humanoid robot as a support assistant to enable a personal aspect as well as have an enhanced impact on the learning experience of students with dyslexia
Detection-Research-Medical	Subudhi et al. (2023)	This paper proposes a ML based ensemble framework to identify neuroatypicality among individuals
Game-Research	Rauschenberger et al. (2022)	An approach for universal screening of dyslexia using ML models with data gathered from a web-based language-independent game
	Tinterri et al. (2024)	This study explores whether trained High-performance AI can facilitate teachers in the creation of boardgame-based learning units
Research-Medical	Muthureka et al. (2023)	An handwritten digit recognition system using optical character recognition (OCR) on dataset of cerebral palsy people's
	García-Ponsoda et al. (2023)	The most used Feature Extraction techniques to classify mental disorders are investigated and presented
Detection-Research	Shalileh et al. (2023)	This paper presents a research to propose a robust artificial intelligence-based solution to identify dyslexia in primary school pupils
	Alqahtani et al. (2023)	This paper discusses the prediction performance of deep learning models for dyslexia summarizing some open challenges
	Vajs et al. (2023)	Bridging the gap between different study designs by developing a ML-based pipeline that was evaluated on two completely different eye-tracking datasets
	Yeguas-Bolívar et al. (2022)	A mobile application integrating VR and AI to collect data quickly and easily, for customizing the supporting methodology for each student

interactive and enjoyable. Games, VR environments, and gamified exercises are explored as methods to increase engagement and retention, providing an alternative approach to traditional learning methods.

Furthermore, recent studies have begun to explore the use of GAI models to develop customized learning materials and conversational agents for learners with dyslexia. These studies suggest a trend toward more dynamic AI-driven educational tools that can offer real-time support, potentially revolutionizing the way educational content is delivered. The reviewed studies highlight the importance of interdisciplinary approaches that combine insights from education, cognitive science, and technology. More

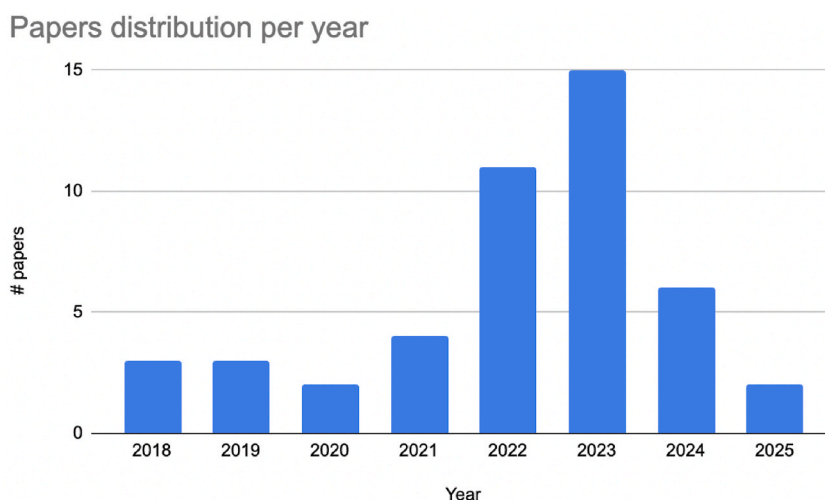


Fig. 8. The Other Papers distribution during time.

experimental research is needed to evaluate the real-world impact of these AI tools on learning outcomes of students with dyslexia, as well as to refine these technologies to better address the specific needs of students with dyslexia.

These general considerations underscore the value of AI as a versatile tool in addressing the various challenges faced by learners with dyslexia. However, the field remains in its early stages, with significant opportunities for research and development to enhance AI's effectiveness in educational and diagnostic contexts.

Despite the systematic approach adopted in this review, several limitations should be acknowledged. First, the categorization of studies into semantic intersections necessarily involved interpretative judgment, which may introduce classification bias. Second, the restriction to English-language publications may have excluded relevant regional or practice-oriented contributions. Third, the heterogeneity of methodologies, evaluation protocols, and outcome measures across studies limits direct comparability and prevents quantitative synthesis. Finally, the rapid emergence of generative AI research—particularly in 2024—suggests that the current evidence base remains preliminary and may evolve significantly in the near future. Future research should prioritize longitudinal and classroom-based evaluations to assess the sustained educational impact of AI tools for learners with dyslexia. Greater theoretical grounding is also needed, integrating frameworks from cognitive science and special education to better understand how AI systems influence cognitive load, motivation, and self-regulation. Additionally, the development of shared benchmarks and standardized reporting practices would improve comparability across studies and strengthen the evidence base. As generative AI continues to evolve rapidly, careful empirical validation will be essential to distinguish short-term novelty effects from meaningful pedagogical innovation.

6.3. Identified research achievements

Throughout our research, many studies showed promise for an easier future in dealing with learning difficulties with dyslexia. We present three research papers that merit discussion because of their implemented and experimented systems with promising results.

In their research, Rajapakse et al. (Rajapakse et al., 2018) developed and tested a system that helps users identify and read the words around them. The prototype tested incorporated a dictionary and a camera that identify complex words and read them out loud in smaller chunks. The system adapts to the disability level of the user. This personalized attention and help to the user that is based on an intelligent knowledge of the language is an example of how AI can help students with dyslexia manage their reading difficulty and daily lives.

The research of Zingoni et al. (Zingoni et al., 2024) showed promising results in reducing the gap between university students with and without dyslexia. A screening tool, DysLexML, was based on data gathered for eye movement while reading text. This is important because it can help to change the teaching modalities depending on the special needs of the students.

Yeguas-Bolívar et al. (Yeguas-Bolívar et al., 2022) used AI methods to identify the most relevant student difficulties and create a personalized help system based on each student. They launched the VRAIlexia project. It consists of a mobile app with two different tools: a virtual reality (VR) tool to collect data from students with dyslexia and an artificial intelligence (AI)-based software to analyze it. This project resulted in a better prediction of the support tools and learning methodologies needed.

7. Open challenges and future trends (RQ2)

In this section, we answer RQ2. The review highlights the diverse applications of AI to support students with dyslexia in education, but several open challenges remain, along with promising future trends.

In fact, given the analysis of the literature discussed in previous sections, several challenges regarding the use of AI to help students

with dyslexia are open. The most promising area was one analyzed in depth, namely the learning aid based on ML and GAI techniques. From what appears in the literature, as of 2024, GAI is taking over from ML techniques, as reflected in the analysis conducted. So in our opinion, the most interesting future trends are those concerning the use of GAI to help students with dyslexia during their learning process.

7.1. Open challenges

From the literature review, a number of challenges are quite clear. The first is limited experimental validation. Although many studies propose AI-based tools and diagnostic methods, there is a lack of extensive experimental validation to evaluate the effectiveness of these tools in real-world educational environments. Experimental studies designed to rigorously assess the impact of AI on learning outcomes for students with dyslexia are essential to advance this field. The second is scalability and accessibility of diagnostic tools. Diagnostic tools that use EEG, eye-tracking, and machine learning models show potential for early detection of dyslexia. However, these technologies often require specialized equipment and controlled environments, making them difficult to implement at scale in typical school settings. Developing accessible and cost-effective diagnostic solutions remains a major challenge. Third, there are ethical and privacy considerations with AI in education. As AI tools become increasingly integrated into educational settings, concerns about data privacy, student consent, and the ethical implications of AI use arise. Ensuring that AI applications respect student privacy and comply with ethical standards is crucial, particularly when dealing with sensitive data from students with learning disabilities. Fourth, there is limited focus on teacher support and training. AI tools are often designed with the end-user, i.e., the students with dyslexia, in mind. However, teachers play a critical role in the effective implementation of these tools. There is a need for AI solutions that include components for teacher support and training to ensure proper integration into the classroom and to empower educators to support students with dyslexia. And finally, we find language and cultural barriers. The prevalence and manifestation of dyslexia can vary between languages and cultures, but most research focuses on English-speaking populations. Expanding research to include diverse languages and cultural contexts is vital for developing AI tools that are inclusive and applicable globally.

7.2. Future Trends

As previously mentioned, the speed of development in AI is astounding, but its application to education is much more measured, and its application to specific learning disabilities even more so. We can nonetheless make several important observations about the future in this particular domain. First, the rise of GAI offers new possibilities for creating personalized educational experiences. Future AI-driven tools could leverage large language models to generate customized reading materials, exercises, and feedback that adapt in real-time to the unique needs of each student with dyslexia, enhancing engagement and comprehension. Second, integrating multiple data sources, such as eye-tracking, EEG, and behavioral analytics, into a unified diagnostic model may improve the accuracy of dyslexia detection. Multi-modal approaches could provide more nuanced insight into the needs of each student, allowing for earlier and more precise interventions. Third, the use of AI-based intelligent tutoring systems, particularly those powered by Natural Language Processing, is a growing trend. These systems could provide interactive, conversational support to students with dyslexia, helping them practice reading and comprehension skills in a low-pressure environment. Conversational agents specifically designed to understand dyslexia-related difficulties could offer a more personalized learning experience. In particular, Chatbots based on GAI in our opinion represent a major development trend. Fourth, future AI applications could include tools specifically designed to support educators in understanding and implementing dyslexia interventions. By providing teachers with information on student progress and suggestions for instructional adjustments, AI could facilitate more effective teaching strategies and foster a supportive learning environment. And finally, addressing the complex needs of students with dyslexia requires collaboration across fields, including AI, education, cognitive science, and psychology. Future research is expected to adopt interdisciplinary frameworks that incorporate insights from multiple fields, leading to more comprehensive solutions that address both educational and cognitive aspects of dyslexia.

These open challenges and trends highlight the potential for AI to transform support for students with dyslexia, while also underscoring the need for careful and ethical development. As AI technologies continue to evolve, interdisciplinary efforts will be essential to ensure that these advancements are accessible, effective, and supportive for all learners.

8. Conclusions

This interdisciplinary review highlights the transformative potential of AI in addressing the educational challenges faced by students with dyslexia, with a special emphasis on the emergent role of GAI. By systematically categorizing and analyzing the literature in semantic fields, such as detection, educational tools, gamification, and teacher training, this study provides a nuanced understanding of how AI innovations intersect with the multifaceted needs of learners with dyslexia.

A central observation is the under-utilization of GAI in this domain, despite its promising capabilities to reshape educational practices. GAI, with its ability to create personalized learning materials, interactive content, and dynamic feedback systems, has significant potential to provide personalized support to students with dyslexia. For example, intelligent chatbots and GAI-powered adaptive platforms can engage learners with interactive and conversational experiences that adjust to their specific difficulties, thus fostering inclusivity and engagement. However, the review reveals that this area remains largely exploratory, with limited rigorous experimentation and real-world validation of such tools.

The interdisciplinary analysis sheds light on the intersections between semantic categories, particularly in the domains of *detection* and *education*, and *help* and *education*. The integration of GAI with traditional machine learning models offers a dual approach: while

traditional models excel in detection and diagnostics through data-driven predictions, GAI can humanize the learning experience by offering interactive, context-sensitive support tailored to individual needs. The role of GAI in teacher training is another area of emerging importance. By equipping educators with AI-powered resources that simplify the preparation and delivery of inclusive education strategies, GAI not only empowers teachers but also bridges the gap between technological innovation and pedagogical practice. This interplay highlights the necessity of a collaborative approach, involving technologists, educators, and cognitive scientists, to design solutions that are practical, ethical, and impactful. Despite its potential, the review highlights several challenges. First, the accessibility and scalability of AI-based tools, including GAI, remain limited, particularly in resource-constrained settings. The reliance on specialized hardware and infrastructure, such as EEG or eye-tracking systems, makes widespread adoption challenging. Second, ethical concerns surrounding data privacy and algorithmic bias must be addressed, particularly when dealing with sensitive information from vulnerable populations such as students with dyslexia. Third, linguistic and cultural diversity in dyslexia manifestations is often overlooked, and most tools are designed for English-speaking contexts. Expanding research to incorporate multilingual and multicultural perspectives is critical to creating truly inclusive solutions. Looking ahead, the study identifies several promising future directions for GAI in this field. These include the development of multi-modal diagnostic tools that combine data sources such as behavioral analytics, language patterns, and physiological signals to improve the accuracy of dyslexia detection. Similarly, AI-driven personalized tutoring systems and intelligent conversational agents tailored for learners with dyslexia can revolutionize how support is delivered. The potential of GAI to dynamically adapt to the progress and needs of the learner introduces a new paradigm in education, one that emphasizes real-time, student-centered interventions. In conclusion, while GAI represents a groundbreaking opportunity to transform the educational landscape for students with dyslexia, its integration into practice requires overcoming significant barriers. Future research must focus on rigorous experimental validation, scalability, and ethical deployment of these technologies. An interdisciplinary approach that unites insights from AI, education, and cognitive science is essential to unlock the full potential of GAI. By addressing these challenges and fostering cross-disciplinary collaboration, the educational experiences of students with dyslexia can be greatly enhanced, offering them the tools and resources to succeed both academically and personally.

References

- Aaron, P. G. (1997). The impending demise of the discrepancy formula. *Review of Educational Research*, 67(4), 461. <https://doi.org/10.2307/1170518>, 461.
- Abd Rahman, H., & Ismail, N. S. (2019). Reading difficulties for pre-school children based on early assessment criteria. In *Proceedings of the second international conference on the future of ASEAN (ICoFA) 2017-Volume 1: Business and social sciences* (pp. 427–434). Springer.
- Ahmad, W., Shokeen, R., & Raj, R. (2025). Artificial intelligence: Solutions in special education. In *Transforming special education through artificial intelligence* (pp. 459–520). IGI Global.
- Alharbi, A. (2022). Analyzing facial expressions and body gestures through multimodal metaphors: An intelligent e-feedback interface. In *International conference on human-computer interaction* (pp. 291–302). Springer.
- Ali, S. A., Nisar, H., Aziz, N. A. A., Fadzil, N. A., Zaber, N. S. M., & Ismail, L. I. (2025). Chapter 8 - Understanding dyslexia and the potential of artificial intelligence in detecting neurocognitive impairment in dyslexia. In A. Subasi, S. M. Qaisar, & H. Nisar (Eds.), *Artificial intelligence and multimodal signal processing in human-machine interaction. Artificial intelligence applications in healthcare and medicine* (pp. 151–170). Academic Press. <https://doi.org/10.1016/B978-0-443-29150-0.00017-2>.
- Aliu, T. V. (2024). Artificial intelligence in special education: A literature review. *Systemic Analytics*, 2(2), 188–199.
- Alluhaidan, A. S., Rashid, M., Aldehim, G., Basheer, S., & Sakri, S. B. (2024). Efficient deep learning model for detecting dyslexia in children from an online gamified test dataset. *Journal of Disability Research*, 3(8), Article 20240099.
- Alqahtani, N. D., Alzahrani, B., & Ramzan, M. S. (2023). Deep learning applications for dyslexia prediction. *Applied Sciences*, 13(5), 2804.
- American Psychiatric Association, DSM-5. (2000). *Task, force: Diagnostic and statistical manual of mental disorders: DSM-4TM* (4th ed.). American Psychiatric Publishing Inc. .
- American Psychiatric Association, DSM-5. (2013). *Task, force: Diagnostic and statistical manual of mental disorders: DSM-5TM* (5th ed.). American Psychiatric Publishing Inc.
- Atkar, G., & Jayaraju, P. (2021). Speech synthesis using generative adversarial network for improving readability of hindi words to recuperate from dyslexia. *Neural Computing & Applications*, 33(15), 9353–9362.
- Barua, P. D., Vicnesh, J., Gururajan, R., Oh, S. L., Palmer, E., Azizan, M. M., Kadri, N. A., & Acharya, U. R. (2022). Artificial intelligence enabled personalised assistive tools to enhance education of children with neurodevelopmental disorders—a review. *International Journal of Environmental Research and Public Health*, 19(3), 1192.
- Benalcázar Chicaiza, D., Barrera G, M. I., Páez Quinde, M. C., & Pilamunga, P. M. E. (2019). M-learning didactic strategy for children diagnosed with dyslexia. In *Digital science* (pp. 143–149). Springer.
- Benmarrakchi, F., Ouherrou, N., Elhammoui, O., & El Kafi, J. (2018). An innovative approach to involve students with learning disabilities in intelligent learning systems. In *International conference on advanced intelligent systems for sustainable development* (pp. 39–50). Springer.
- Bhuvan, B., Karthikeyan, I. P., & Rajesh, B. (2024). Can chatgpt empower people with dyslexia? Disability and Rehabilitation. *Assistive Technology*, 19(5), 2131–2132. <https://doi.org/10.1080/17483107.2023.2256805>
- Bu, X., Wang, Y., & Du Yawen, W. (2023). Application of machine learning in early screening of children with dyslexia. *Advances in Psychological Science*, 31(11), 2092.
- Caute, A., Cruice, M., Marshall, J., Monnelly, K., Wilson, S., & Woolf, C. (2018). Assistive technology approaches to reading therapy for people with acquired dyslexia. *Aphasiology*, 32(sup1), 40–42.
- Chen, J. J., & Perez, C. (2023). Enhancing assessment and personalized learning through artificial intelligence. *Childhood Education*, 99(6), 72–79.
- Christodoulou, J. A., Del Tufo, S. N., Lymberis, J., Saxler, P. K., Ghosh, S. S., Triantafyllou, C., Whitfield-Gabrieli, S., & Gabrieli, J. D. (2014). Brain bases of reading fluency in typical reading and impaired fluency in dyslexia. *PLoS One*, 9(10), Article 100552. <https://doi.org/10.1371/journal.pone.0100552>
- Dabaghi, K., D'Urso, S., & Sciarrone, F. (2024). Artificial intelligence and learning of students with dyslexia: A brief review. In Z. Kubincová, T. Hao, N. Capuano, M. Temperini, S. Ge, Y. Mu, P. Fantozzi, & J. Yang (Eds.), *Emerging technologies for education* (pp. 155–169). Singapore: Springer.
- De Marco, V., Sciarrone, F., & Temperini, M. (2024). Tutorchat: A chatbot for the support to dyslexic learner's activity through generative ai. In *Proceedings of the 24th IEEE international conference on advanced learning technologies, ICALT 2024*.
- Della Ventura, M. (2019). Exploring the impact of artificial intelligence in music education to enhance the dyslexic student's skills. In *Learning Technology for Education Challenges: 8th International Workshop, LTEC 2019, Zamora, Spain, July 15–18, 2019, Proceedings 8* (pp. 14–22). Springer.
- Dobson Waters, S., & Torgerson, C. J. (2021). Dyslexia in higher education: A systematic review of interventions used to promote learning. *Journal of Further and Higher Education*, 45(2), 226–256.
- D'Urso, S., & Sciarrone, F. (2024). Ai4la: An intelligent chatbot for supporting students with dyslexia, based on generative ai. In *International conference on intelligent tutoring systems* (pp. 369–377). Springer.

- El Kah, A., & Lakhouaja, A. (2018). Developing effective educative games for Arabic children primarily dyslexics. *Education and Information Technologies*, 23(6), 2911–2930. <https://doi.org/10.1007/s10639-018-9750-2>
- García-Ponsoda, S., García-Carrasco, J., Teruel, M. A., Maté, A., & Trujillo, J. (2023). Feature engineering of eeg applied to mental disorders: A systematic mapping study. *Applied Intelligence*, 53(20), 23203–23243.
- Gellert, A. S., & Elbro, C. (2018). Predicting reading disabilities using dynamic assessment of decoding before and after the onset of reading instruction: A longitudinal study from kindergarten through grade 2. *Annals of Dyslexia*, 68, 126–144.
- Gilbert, B., Stubblefield, J., Qualls, J., Huang, X., Pait, A., Yanowitz, K., Hays, A., Richmond, E., Parker, L., & Washington, T. (2023). Dyslexia and ai: The use of artificial intelligence to identify and create font to improve reading ability of individuals with dyslexia. In *Society for information technology & teacher education international conference* (pp. 856–865). Association for the Advancement of Computing in Education (AACE).
- Goodman, S. M., Buehler, E., Clary, P., Coenen, A., Donsbach, A., Horne, T. N., Lahav, M., MacDonald, R., Michaels, R. B., Narayanan, A., Pushkarna, M., Riley, J., Santana, A., Shi, L., Sweeney, R., Weaver, P., Yuan, A., & Morris, M. R. (2024). Lampost: Ai writing assistance for adults with dyslexia using large language models. *Communications of the ACM*.
- Grover, D. (2024). Next-generation education: The impact of generative ai on learning. *Journal of Informatics Education and Research*, 4(2).
- Guo, J. V., Francis, P. A., & Tan, G. (2023). D-pen—personalised practice with assistive pen for young dyslexic learners using machine learning. In *IRC-SET 2022: Proceedings of the 8th IRC conference on science, engineering and technology, August 2022, Singapore* (pp. 209–243). Springer.
- Gupte, N., Patel, M., Pen, T., & Kurhade, S. (2023). Early detection of adhd and dyslexia from eeg signals. In *2023 IEEE 8th international conference for convergence in technology (I2CT)* (pp. 1–5). IEEE.
- Habib, S., Vogel, T., Anli, X., & Thorne, E. (2024). How does generative artificial intelligence impact student creativity? *Journal of Creativity*, 34(1), Article 100072.
- Hauwaert, H., Ghesquiere, P., Tordoir, J., & Thomson, J. (2020). Karaton: An example of ai integration within a literacy app. In *Computers helping people with special needs: 17th international conference, ICCHP 2020, Lecco, Italy, September 9–11, 2020, proceedings, part I 17* (pp. 91–96). Springer.
- Ileri, R., Latifoglu, F., & Demirci, E. (2022). A novel approach for detection of dyslexia using convolutional neural network with eeg signals. *Medical, & Biological Engineering & Computing*, 60(11), 3041–3055.
- Jaramillo-Alcázar, A., Venegas, E., Criollo-C, S., & Luján-Mora, S. (2021). An approach to accessible serious games for people with dyslexia. *Sustainability*, 13(5), 2507.
- Kaisar, S. (2020). Developmental dyslexia detection using machine learning techniques: A survey. *ICT Express*, 6(3), 181–184.
- Kaisar, S., & Chowdhury, A. (2022). Integrating oversampling and ensemble-based machine learning techniques for an imbalanced dataset in dyslexia screening tests. *ICT Express*, 8(4), 563–568.
- Kaluri, R., Mahmud, M., Gadekallu, T. R., Rajput, D. S., & Lakshmana, K. (2024). *Applied assistive technologies and informatics for students with disabilities*. Springer.
- Kamarozaman, M. I. Z. B. I., Hafizul, A. N. b. Y. M., & Bin, B. A. M. M. (2023). Speed reading tool powered by artificial intelligence for students with adhd, dyslexia, or short attention span. *arXiv preprint arXiv:2307.14544*.
- Kara, A. (2021). Covid-19 pandemic and possible trends into the future of higher education: A review. *Journal of Education and Educational Development*, 8(1).
- Kearns, D. M., Hancock, R., Hoeff, F., Pugh, K. R., & Frost, S. J. (2019). The neurobiology of dyslexia. *Teaching Exceptional Children*, 51(3), 175–188. <https://doi.org/10.1177/0040059918820051>
- Khaleghi, A., Aghaei, Z., & Behnamghader, M. (2022). Developing two game-based interventions for dyslexia therapeutic interventions using gamification and serious games approaches entertainment computing journal. *Entertainment Computing*, 42, Article 100482.
- Khan, R. U., Cheng, J. L. A., & Bee, O. Y. (2018). Machine learning and dyslexia: Diagnostic and classification system (dcs) for kids with learning disabilities. *International Journal of Engineering & Technology*, 7(3.18), 97–100.
- Kharbat, F. F., Alshawabkeh, A., & Woolsey, M. L. (2021). Identifying gaps in using artificial intelligence to support students with intellectual disabilities from education and health perspectives. *Aslib Journal of Information Management*, 73(1), 101–128.
- Khazanchi, R., & Khazanchi, P. (2024). Generative ai to improve special education teacher preparation for inclusive classrooms. In M. Searson, E. Langran, & J. Trumble (Eds.), *Exploring new Horizons: Generative artificial intelligence and teacher education* (pp. 159–177). AACE.
- Knight, C. (2018). What is dyslexia? An exploration of the relationship between teachers' understandings of dyslexia and their training experiences. *Dyslexia*, 24(3), 207–219. <https://doi.org/10.1002/dys.1593>
- Kohli, M., & Prasad, T. (2010). Identifying dyslexic students by using artificial neural networks. In *1. Proceedings of the World Congress on Engineering* (pp. 1–4).
- Leij, A. (2013). Dyslexia and early intervention: What did we learn from the dutch dyslexia programme? *Dyslexia*, 19(4), 241–255. <https://doi.org/10.1002/dys.1466>
- Lerga, R., Candric, S., & Jakupovic, A. (2021). A review on assistive technologies for students with dyslexia. *CSEDU*, (2), 64–72.
- Li, N. (2024). Generative reader: Exploring the use of generative ai in enhancing reading experiences for children with dyslexia. In *INTED2024 proceedings* (pp. 8023–8027). IATED.
- Lillywhite, A., & Wolbring, G. (2023). Coverage of well-being within artificial intelligence, machine learning and robotics academic literature: The case of disabled people. *AI & Society*, 1–19.
- Lopez, D., & Rangappa, S. (2024). The impact of ai on early intervention and education of children with dyslexia. In *EDULEARN24 proceedings* (pp. 6777–6786). IATED.
- Ma, Q. (2023). Exploring vr games for enhancing reading fluency for students with reading disabilities. *Journal of Education, Humanities and Social Sciences*, 22, 244–248.
- MacArthur, C. A. (2009). Reflections on research on writing and technology for struggling writers. *Learning Disabilities Research & Practice*, 24(2), 93–103.
- Mateos-Sanchez, M., Melo, A. C., Blanco, L. S., & García, A. M. F. (2022). Chatbot, as educational and inclusive tool for people with intellectual disabilities. *Sustainability*, 14(3), 1520.
- Mayilyan, A. (2024). Integrating generative ai in stem education: A pathway to inclusive and enhanced learning. *Education in the 21st Century*, 11(1), 93–101.
- Mazefsky, C., McPartland, J., Gastgeb, H., & Minshew, N. (2013). Brief report: Comparability of dsm-iv and dsm-5 asd research samples. *Journal of Autism and Developmental Disorders*, 43, 1236–1242.
- Mcvey, S.-M., Chew, E., & Carroll, F. (2022). Educational robotics and dyslexia: Investigating how reinforcement learning in robotics can be used to help support students with dyslexia. In *International conference on technological ecosystems for enhancing multicultural* (pp. 43–49). Springer.
- Minoofam, S. A. H., Bastanfard, A., & Keyvanpour, M. R. (2022). Ralf: An adaptive reinforcement learning framework for teaching dyslexic students. *Multimedia Tools and Applications*, 81(5), 6389–6412.
- Miranda Floriano, H., Bonacin, R., & Franco Rosa, F. (2022). A user profile based method for usability assessment of distance learning systems. In *International conference on human-computer interaction* (pp. 275–288). Springer.
- Mitra, S., Lakshmi, D., & Govindaraj, V. (2023). Data analysis and machine learning in ai-assisted special education for students with exceptional needs. In *AI-Assisted special education for students with exceptional needs* (pp. 67–109). IGI Global.
- Morciano, G., Llergo, J. M. A., Zingoni, A., Bolívar, E. Y., Taborri, J., & Calabrò, G. (2024). Use of recommendation models to provide support to dyslexic students. *Expert Systems with Applications*, 249, Article 123738.
- Mortimore, T., & Crozier, W. R. (2006). Dyslexia and difficulties with study skills in higher education. *Studies in Higher Education*, 31(2), 235–251. <https://doi.org/10.1080/03075070600572173>
- Muthureka, K., Srinivasulu Reddy, U., & Janet, B. (2023). An improved customized cnn model for adaptive recognition of cerebral palsy people's handwritten digits in assessment. *International Journal of Multimedia Information Retrieval*, 12(2), 23.
- Nair, B. M., et al. (2023). The efficacy of artificial intelligence-driven immersive reader for dyslexic students in special schools: A case study. *Journal of English Language Teaching*, 65(5), 3–8.
- Ogunlade, B. O., Babatunde, R. A., & Fakuade, O. V. (2023). Using conversational artificial intelligence tool in rebuilding reading and spelling on dyslexic learners in nigerian schools. *Uluslararası Özel Amaçlar için İngilizce Dergisi*, 1(1), 1–11.
- Ouhrourou, N., Elhammoumi, O., Benmarakchi, F., & El Kafi, J. (2018). A heuristic evaluation of an educational game for children with dyslexia. In *2018 IEEE 5th international congress on information science and technology (CIST)* (pp. 386–390). IEEE.

- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., et al. (2021). The prisma 2020 statement: An updated guideline for reporting systematic reviews. *Bmj*, 372.
- Pait, A., Qualls, J., Gilbert, B. B., Hays, A., Yanowitz, K., Huang, X., Carter, G., & Antici, J. (2024). Data analysis of computer-generated font and student reading performance: Dyslexia and ai-next steps. In *Society for information technology & teacher education international conference* (pp. 1960–1964). Association for the Advancement of Computing in Education (AACE).
- Praveena, K., Mahalakshmi, R., Manjunath, C., & Dakhole, D. K. (2024). Autism, adhd and dyslexia disorder comorbidity: An enhanced study on education for children through artificial intelligence-enabled personalized assistive tools. In *Handbook of AI-Based models in healthcare and medicine* (pp. 437–450). CRC Press.
- Quinones, M. (2020). The future of literature: Neuromarketing and audio books: A new opportunity for content marketing. In *Academy of marketing science annual conference* (pp. 95–105). Springer.
- Rajapakse, S., Polwattage, D., Guruge, U., Jayathilaka, I., Edirisinghe, T., & Thelijjagoda, S. (2018). Alexza: A mobile application for dyslexics utilizing artificial intelligence and machine learning concepts. In *2018 3rd international conference on information technology research (ICITR)* (pp. 1–6). <https://doi.org/10.1109/ICITR.2018.8736130>
- Rauschenberger, M., Baeza-Yates, R., & Rello, L. (2022). A universal screening tool for dyslexia by a web-game and machine learning. *Frontiers in Computer Science Human-Media Interaction*, 3, 2021.
- Reis, A., Castro, S. L., Petersson, K. M., & Reis, A. (2020). Reading and reading-related skills in adults with dyslexia: A review. *Annals of Dyslexia*, 70(3), 309–327. <https://doi.org/10.1007/s11881-020-00196-0>
- Richardson, U., & Lyytinen, H. (2014). The graphogame method: The theoretical and methodological background of the technology-enhanced learning environment for learning to read. *Human Technology*, 10(1).
- Rose, D. H., & Meyer, A. (2002). *Teaching every student in the digital age: Universal design for learning*. ERIC.
- Ruberto, N., Daigle, D., & Ammar, A. (2016). The spelling strategies of francophone dyslexic students. *Reading and Writing*, 29(4), 659–681. <https://doi.org/10.1007/s11145-015-9620-x>
- Selwyn, N. (2021). *Education and technology: Key issues and debates*. Bloomsbury Publishing.
- Shalileh, S., Ignatov, D., Lopukhina, A., & Dragoy, O. (2023). Identifying dyslexia in school pupils from eye movement and demographic data using artificial intelligence. *PLoS One*, 18(11), Article 0292047.
- Shardlow, M., Sellar, S., & Rousell, D. (2022). Collaborative augmentation and simplification of text (coast): Pedagogical applications of natural language processing in digital learning environments. *Learning Environments Research*, 1–23.
- Sharif, M. S., & Elmedany, W. (2022). A proposed machine learning based approach to support students with learning difficulties in the post-pandemic norm. In *2022 IEEE global engineering education conference (EDUCON)* (pp. 1988–1993). IEEE.
- Shaywitz, S. E. (2012). *Overcoming dyslexia : A new and complete science-based program for reading problems at any level*. A.A. Knopf, New York.
- Siegel, L. S. (2006). Perspectives on dyslexia. *Paediatrics and Child Health*, 11(9), 581–587. <https://doi.org/10.1093/pch/11.9.581>
- Simmons, F. R., & Singleton, C. (2008). Do weak phonological representations impact on arithmetic development? A review of research into arithmetic and dyslexia. *Dyslexia*, 14(2), 77–94. <https://doi.org/10.1002/dys.341>
- Smith, C., & Hattigh, M. (2020). Assistive technologies for students with dyslexia: A systematic literature review. In *Innovative Technologies and Learning: Third International Conference, ICITL 2020, Porto, Portugal, November 23–25, 2020, Proceedings 3* (pp. 504–513). Springer.
- Stanley, J., Brink, R. T., Valiton, A., Bostic, T., & Scollan, B. (2022). Chatbot accessibility guidance: A review and way forward. In , 3. *Proceedings of sixth international congress on information and communication technology: ICICT 2021, London* (pp. 919–942). Springer.
- Subramani, S., Jerish S. A., C. S., M. S., T. M., & M. M. (2024). Learn buddy – Transforming education with generative ai. In *2024 international conference on computing and data science (ICCCDS)* (pp. 1–5). <https://doi.org/10.1109/ICCCDS60734.2024.10560378>
- Subramanyam, A., Jyrwa, S., Bansinghani, J. M., Dadhakar, S. J., Dhinra, T. V., Ramchandani, U. R., & Sengupta, S. (2019). Dyscalculia detection using machine learning. In *Pattern recognition and machine intelligence: 8th international conference, PReMI 2019, Tezpur, India, December 17–20, 2019, proceedings, part I* (pp. 111–120). Springer.
- Subudhi, A., Hasan, N., & Nene, M. J. (2023). Deen: Deep ensemble framework for neuroatypicality classification. In *2023 3rd international conference on intelligent technologies (CONIT)* (pp. 1–6). IEEE.
- Tayal, A. (2021). An interactive alphabet and number learning system using opencv and cnn. In *2021 Asian conference on innovation in technology (ASIANCON)* (pp. 1–4). IEEE.
- Tejero, P., Pi-Ruano, M., & Roca, J. (2020). Better read it to me: Benefits of audio versions of variable message signs in drivers with dyslexia. *Annals of Dyslexia*, 70(3), 295–312.
- Tinterri, A., Di Padova, M., Palladino, F., Vignoli, G., Dipace, A., et al. (2024). Ai in board game-based learning. In *Ceur workshop proceedings*.
- Tunmer, W., & Greaney, K. (2010). Defining dyslexia. *Journal of Learning Disabilities*, 43(3), 229–243. <https://doi.org/10.1177/0022219409345009>
- Vajs, I. A., Kvaščev, G. S., Papić, T. M., & Janković, M. M. (2023). Eye-tracking image encoding: Autoencoders for the crossing of language boundaries in developmental dyslexia detection. *IEEE Access*, 11, 3024–3033.
- Vandermosten, M., Hoeff, F., & Norton, E. S. (2016). Integrating mri brain imaging studies of pre-reading children with current theories of developmental dyslexia: A review and quantitative meta-analysis. *Current Opinion in Behavioral Sciences*, 10, 155–161.
- Vukovic, R. K., Lesaux, N. K., & Siegel, L. S. (2010). The mathematics skills of children with reading difficulties. *Learning and Individual Differences*, 20(6), 639–643. <https://doi.org/10.1016/j.lindif.2010.08.004>
- Wadhera, T., & Mahmud, M. (2022). Influences of social learning in individual perception and decision making in people with autism: A computational approach. In *International conference on brain informatics* (pp. 50–61). Springer.
- Wang, M., Muthu, B., & Sivaparthipan, C. (2021). Smart assistance to dyslexia students using artificial intelligence based augmentative alternative communication. *International Journal of Speech Technology*, 1–11.
- Wang, M., Muthu, B., & Sivaparthipan, C. (2022). Smart assistance to dyslexia students using artificial intelligence based augmentative alternative communication. *International Journal of Speech Technology*, 25(2), 343–353.
- Xu, L., Luo, L., & Qin, R. (2021). Application of intelligent learning environment in the education of people with dyslexia. In *2021 2nd international conference on information science and education (ICISE-IE)* (pp. 1657–1660). IEEE.
- Yeguas-Bolívar, E., Alcalde-Llargo, J. M., Aparicio-Martínez, P., Taborri, J., Zingoni, A., & Pinzi, S. (2022). Determining the difficulties of students with dyslexia via virtual reality and artificial intelligence: An exploratory analysis. In *2022 IEEE international conference on metrology for extended reality, artificial intelligence and neural engineering (MetroXRINE)* (pp. 585–590). IEEE.
- Zavaraki, E. Z. (2024). Artificial intelligence for people with special educational needs. *IntechOpen*.
- Zhai, X., & Krajcik, J. (2023). *Uses of artificial intelligence in STEM education*. Oxford University Press.
- Zingoni, A., Taborri, J., & Calabrò, G. (2024). A machine learning-based classification model to support university students with dyslexia with personalized tools and strategies. *Scientific Reports*, 14(1), 273.